بِسْمِ ٱللَّهِ ٱلرَّحْمَٰنِ ٱلرَّحِيمِ

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

Department of Computer Science & Engineering (CSE)

BSc in CSE, Mid-Term Examination, Spring 2025

Course Code: **CSE-3527** Course Title: **Compiler**

Full Marks: **30** Time: **1 Hour 30 minutes**

*[Answer **all** the following questions. Some questions may have an option.*
(Figures at right margin illustrate the marks (M), course objectives (COs), and bloom's taxonomy (DL))]

No.	Questions	M	COs	DL
1. a)	Define the compiler with a block diagram that illustrating its front-end and back-end phases. Elucidate the analysis of the source program with its compiling phases.	4 (2+2)	CO1	U
	Or			
	How is an executable target program generated from a given source program? Explain it using a language processing system with an appropriate diagram. What basic tools are needed for compiler construction in a language processing system?	4 (1+2+1)	CO1	An
b)	Translate the following C programming statement step by step through the compiler phases. Draw the diagram of the compiler's phases with output translating annotations at each phase and generate the symbol table for the following statement. `a = x + y * r - w / 4;` [where the values of `a, r, and w` are fractional]	6 (2+3+1)	CO1	An
2. a)	Write is the purpose of context-free grammar (CFG) in compiler design. Define and emphasize the concept of ambiguity in CFG. How does it impact a compiler design? Consider the following CFG G_1: $Exp \rightarrow Exp\ Op\ Exp$; $Exp \rightarrow (Exp)$; $Op \rightarrow + \mid - \mid * \mid /$; $Exp \rightarrow int$. Whether the above grammar is ambiguous or not. Justify your answer using its input string "`int * (int + int)`". Construct the syntax tree for the input string "`int + int * (int / int)`".	6 (3+3)	CO2	Ap
	Or			
	Define synthesized attributes? Consider the following CFG G_2: and G_3	6 (1+2+3)	CO2	An

G_2:

$S \rightarrow Ab \mid aB$
$A \rightarrow aA \mid \varepsilon$
$B \rightarrow Bb \mid \varepsilon$

G_3:

$Seq \rightarrow Seq\ Instr \mid \mathbf{begain}$
$Instr \rightarrow \mathbf{east} \mid \mathbf{north} \mid \mathbf{west} \mid \mathbf{south}$

i) Write the language definition of the above grammar G_2 with $\Sigma = \{a, b\}$ and generate the leftmost and rightmost derivation for the input string "**aaab**"

ii) For a robot position tracking system CFG G3, suppose a robot can instructed to move east, north, west, or south from the current position. Position can be marked by a coordinate pair (x, y) with an initial position (0,0) at the **beginning**. Draw the tracking position of the robot and give the leftmost derivation and parse tree for the input string "**begin west south east east north north**"

Symbol
Exception handling

b) Describe the roles of lexical analyzer and parser in a compiler with the appropriate block diagram. Identify the tokens, lexemes and patterns of the following C programming code segment in a table [please avoid the redundant entry of tokens]: — 4 (2+2) — CO2 U

```
int c, n=5, fact = 1;
printf("Enter a integer number\n");
for(c = 1; c <= n; c++)
    fact = fact * c;
printf("Factorial of %d = %d\n", n, fact);
```

3. a) Write the regular expression and draw the nondeterministic finite automata (NFA) transition diagram to define the number of C programming language including fractional and exponential parts. Construct the $\epsilon - NFA$ for regular expression "`(0|1)*011`" and convert it into the deterministic finite automata (DFA) using ϵ-closure. — 6 (2+4) — CO2 An

b) Define the different types of grammar with emphasizing the basic production rules of the grammar. Identify the regular grammars, right-linear grammars, left-linear grammars, context free grammars, context sensitive grammars and useless non-terminals in the following grammars. [Some grammar may fit more than one category]. — 4 (2+2) — CO2 Ap

G4:	$G5$:	$G6$:	$G7$:
S → baB	$S \rightarrow aX\,\vert bY$	$S \rightarrow aA\vert\, \varepsilon$	$S \rightarrow Ma\vert Mb$
aB → aA	$X \rightarrow aY$	$A \rightarrow Ab\vert a$	$M \rightarrow Nb\vert d$
A → bA|b	$Y \rightarrow b$	$C \rightarrow$ c	$N \rightarrow$ c

END

L→0|..|9

L+.L*

0.5

0

3

9.5

t4 = id4 of (id5)/4

t5 = id4 -

~~t4 = (id1~~

t4 = id3/4

t5 = id4 - t4

t6 = id3 + t5

id1 = id2 + t6

SR

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG
Department of Computer Science & Engineering (CSE)

BSc in CSE, Mid-Term Examination, Spring 2024
Course Code: **CSE-3527** Course Title: **Compiler**

Full Marks: **30** Time: **1 Hour 30 minutes**

[Answer all of the following questions. Some questions may have an option.
(Figures at right margin illustrate the marks (M), course objectives (COs), and bloom's taxonomy (DL))]

No.	Questions	M	COs	DL
1. a)	Elucidate the block diagram to define a tool to convert an input source program into an output target program and contrast it with the interpreter. What is the significant role of this tool in a language processing system, and illustrate it with an appropriate figure of a language processing system?	4 (2+2)	CO1	U
b)	Translate the following C programming assignment statement step by step in compiler phases. Draw the diagram of the compiler's phases with output annotation of each phase and symbols table. `net_salary = basic + house_rent - basic * tax_rate;` [where the value of `tax_rate` is fractional]	6 (4+2)	CO1	An

2. a) Illustrate the concept of ambiguity and how it impacts on a compiler design? Consider the following CFG G_1: **5 (2+3) CO2 Ap**

$$Exp \rightarrow Exp + Exp \mid Exp - Exp$$
$$Exp \rightarrow Exp * Exp \mid Exp \,/\, Exp$$
$$Exp \rightarrow 0|1|2|3|4|5|6|7|8|9$$
$$Exp \rightarrow id$$

Justify whether the above grammar is ambiguous using its input string **"6-5+2*1/id"**. If it is ambiguous, convert it into the unambiguous grammar and construct the syntax tree for the same input string.

Or **CO2 An**

What do you mean by synthesized attributes? Consider the following CFG G_2: **5 (1+2+2)**

$$E \rightarrow E + T \mid E - T$$
$$E \rightarrow T$$
$$T \rightarrow 0|1|2|3|4|5|6|7|8|9$$

i) Write the syntax directed definition using semantic rules for the above CFG G_2 and Create an annotated parse tree for the input string: *"9-5+2-1"*

ii) Generate leftmost and rightmost derivation for the input string: *"5+1-4+8"*

b) Illustrate the roles of lexical analyzer in a compiler with schematic diagram. What is the main advantage of separating the analysis phase of compiling into lexical analysis and parsing? Identify the tokens, lexemes and patterns of the following C programming function in a table: **5 (2+1+2) CO2 U**

```
long factorial (int n) {
    if (n==1)
        return 1;
    else
        return n*factorial(n-1);
}
```

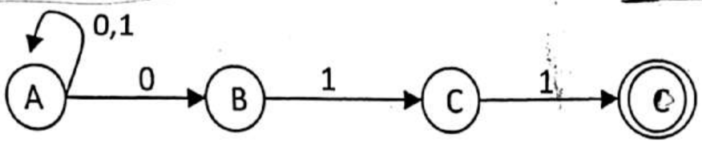
$T_1 = id_3 * T_1$
$T_2 = id_2 + T_2$
$T_3 = id_2 + T_2$
$id_1 = T$

$E \rightarrow E * E$
$\rightarrow E * E / E$
$\rightarrow E + E * E / E$
$\rightarrow E - E + E * E / E$
$\rightarrow 6 - 5 + 2 * 1 / id$

3. a) Why finite automata is important in a compiler construction? Write the regular expression to define the identifier of C programming language and construct the $\epsilon - NFA$ and DFA using that regular expression. 5 (1+1+2+1) CO2 An

Or

Contrast DFA with NFA. Construct the equivalent NFA for the regular expression RE1: ***(a|b) *abb(a|b)*** directly and Convert the following NFA to DFA. 5 (1+2+2) CO2 An



b) Define syntax directed translation of *if-else* conditional statement of C programming language using appropriate grammar. What is the main factor to identify the types of grammar? Identify the regular grammars, right-linear grammars, left-linear grammars, context free grammars, context sensitive grammars and useless non-terminals in the following grammars. 5 (2+3) CO2 Ap

[Some grammar may fit more than one category].

G3:	G4:	G5:	G6:
$S \rightarrow aM$	$S \rightarrow aX \mid \varepsilon$	$A \rightarrow aA \mid \varepsilon$	$S \rightarrow Aa \mid Ab$Ty|
$M \rightarrow Pb$	$X \rightarrow aY$	$B \rightarrow Ab \mid b$	$B \rightarrow Cb \mid Dd$
$Ma \rightarrow bPc$	$Y \rightarrow b$	$C \rightarrow c$	$C \rightarrow c$
$Pc \rightarrow ac$			

$A \rightarrow Aa$

$A \rightarrow Ba$

$A \rightarrow \alpha B$

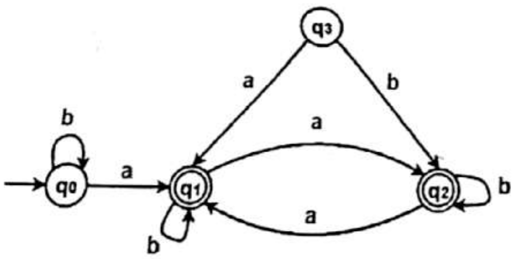
CFG

CSG

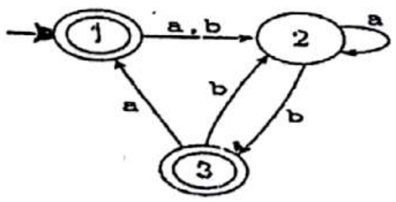
END

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE Midterm Examination, Autumn 2023
Course Code: CSE-3527 **Course Title:** Compiler
Total marks: 30 **Time:** 90 minutes

			Marks	CO	C
1.	a.	What is bytecode? How do Java language processors work?	2	CO1	C2
	b.	How is an executable target program generated? Explain it using a language processing system.	3	CO1	C2
	c.	Translate the following assignment statement step by step using compiler phases: $z = a * b + c/4;$ [$z, a, b,$ and c contain floating point values]	4	CO1	C3
2.	a.	Consider the CFG: $S \to +SS \mid *SS \mid S \mid a$ 1. Show that: $aa^{+}a^{*}$ can be generated by the grammar. +a*aa 2. Draw the parse tree	2	CO1	C1 C2
	b.	Consider the grammar: $P \to aPbP \mid bPaP \mid \epsilon$ Consider $ababab$ as the string to check the ambiguity of the given grammar.	4	CO1	C3
	c.	Construct the regular expressions and provide two accepted strings for each. **(Any two)**: 1) $\{w \mid w: \text{starts with a 'b' and has at least one } a\}$ 2) $\{w \mid w: \text{does not contain the substring } abab\}$ 3) $\{w \mid w: \text{set of strings consisting of even number of } a\text{'s followed by } b\}$ 4) $\{w \mid w: \text{set of strings of 1's and 0's in even length}\}$	4	CO2	C3
3.	a.	Minimize the following DFA:	4	CO2	C3



			Marks	CO	C
	b.	Convert the following regular expression directly to its equivalent NFA: $abb\,(a\|b)*ba$ **OR,** $(a\|b)^{*}a(a\|b)(a\|b)$	2	CO2	C3
	c.	Convert the following NFA to DFA:	4	CO1	C3



CamScanner

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Midterm Examination, Autumn 2022

Course Code: CSE 3527 Course Title: Compiler

Total marks: 30

*Time: **1 Hour and 30 minutes***

			CO	DL
1.				
a)	Translate the following assignment statement step by step using compiler phases: z = c/d + a*b, here z, c, d are integer and a, b are floats.	5	CO1	C3

b)	Write the differences between lexical analyzer and parser. Identify the lexemes and their corresponding tokens and non-tokens from the following C code:
	#include<stdio.h>
	#define a 10
	int sum(float x, float y)
	{
	float m;
	// this will add 2 numbers
	sum= x+y;
	if(x>y)
	printf("%d", sum);
	else
	return sum;
	}
		5	CO1	C2, C3

2.				
a)	What is context sensitive grammar? Identify the useless non terminals, Left Linear Grammar, Right Linear Grammar from the following grammar? S -> ABd/a A -> eBC/b B -> aB/C C -> aC/B Or, When do we call a grammar ambiguous? Check whether the given grammar G is ambiguous or not for the string "a(a)aa". $A \rightarrow AA$ $A \rightarrow (A)$ $A \rightarrow a$	4	CO3	C1, C2
b)	Construct CFG for the language of all non-Palindromes. Or, Construct a CFG for the language $L = 0^n1^{4n}$ where $n>=1$.	2	CO3	C3
c)	Write the regular expressions for the following languages: i. All strings of a's and b's with an even number of a's and an odd number of b's. ii. All strings of a's and b's that do not contain the substring abb.	4	CO2	C3
3.				
a)	Find the regular expression for the following FA:	4	CO2	C3

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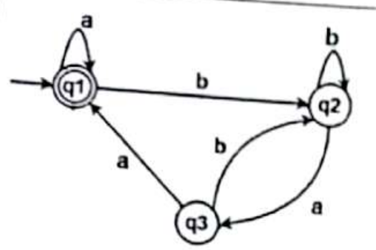
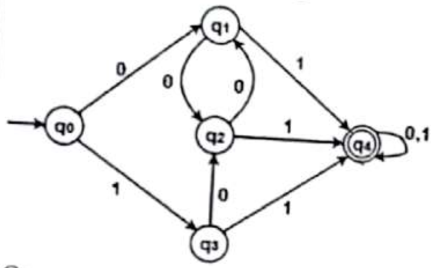
<tr><td></td><td>
</td><td></td><td></td><td></td></tr>
<tr><td>a)</td><td>Minimize the following DFA:


Or,
Convert the regular expression R= abb(a|b)*(a|b) directly to its equivalent NFA</td><td>4</td><td>CO2</td><td>C3</td></tr>
<tr><td>b)</td><td>Describe the languages denoted by the following regular expressions:
i. a(a|b)*a
ii. a*ba*ba*ba*</td><td>2</td><td>CO2</td><td>C3</td></tr>
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International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE, Midterm Exam, 5th Semester, Spring 2022

Course Code: CSE 3527 | **Course Title: Compiler**

Total marks: **30** | Time: **1** hours **30** minutes

[Answer all the questions; in some questions, there are options; Figures in the right-hand margin indicate full marks.]

1. **a.** Specify each phaseof a compiler explicitly with corresponding input and output while processing the following statement: **5**

Expression := Term * factor + 128

b. Consider the grammar: **5**

$$
\begin{aligned}
stmt &\rightarrow if\text{-}stmt \mid \texttt{other} \\
if\text{-}stmt &\rightarrow \texttt{if}\ (\ exp\ )\ stmt \\
&\mid \ \texttt{if}\ (\ exp\ )\ stmt\ \texttt{else}\ stmt \\
exp &\rightarrow \texttt{0} \mid \texttt{1}
\end{aligned}
$$

i. Now, show that the grammar is ambiguous by constructing two parse trees for the following string:
if (0) if (1) other else other

ii. Rewrite the grammar so that it becomes unambiguous.

2. **a.** Consider the following grammar: **3**

E→ E or T | T
T →T and F | F
F→ not F | (E) | true | false

Now, construct a parse tree for the following sequence:

not (true or false)

b. Suppose you are considering DFA from regular expression. Now, compute the ***followpositions*** for the following regular expression: **7**
(a|b)*ab*ab

OR

Convert the regular expression **R=(1|0)*1** to its equivalent NFA using Thompson construction.

3. **a.** i. What are the advantages of separating the analysis phase of compiling into lexical analysis and parsing? **5**

ii. Why context free grammar is preferred over regular expressions in parsing?

b. Convert the regular expression **R=(a|b)*abb(a|b)** directly to its equivalent DFA. **5**

OR

Prove that **0*(10*)*** is equivalent to **(1*0)*1***